Unit 1 · Session 2 · Thu, Oct 22, 2026

Introduction to Pharmacology — Session 2

5 parts · every fact cited to its slide or textbook page. About 150 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

50 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

30 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

45 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Metabolism inactivates drugs, activates prodrugs, or makes active or **toxic metabolites** (e.g., NAPQI); **Phase I** (CYP450 oxidation, reduction, hydrolysis) functionalizes and **Phase II** conjugates for excretion. S4; S4 (notes); S5

2. A **CYP inhibitor raises** a substrate's level (toxicity) and an **inducer lowers** it (therapeutic failure) — always check CYP450 interactions before adding a drug. S6; S7 (notes); S40

3. The **kidney** is the main elimination route — filtration of free drug, active secretion, reabsorption of lipophilic un-ionized drug — and less renal function means less clearance and more exposure. S13; S14; S15

4. **Half-life** (time for concentration to fall 50%) depends on clearance and Vd: **~4–5 half-lives** to steady state and **~97%** eliminated after ~5. S19; S23

5. **Loading doses** (based on Vd) reach the target immediately; **maintenance doses** (based on clearance) keep levels in range; in CKD adjust dose and/or interval by eGFR/CrCl. S23; S23 (notes); S28

6. **Pharmacogenomic** variants (CYP2D6, CYP2C19, TPMT, HLA-B*5801) change drug response, but genotype is one input in a checklist of indication, contraindications, organ function, interactions, patient factors, and dose/route/monitoring. S33; S34; S39

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO1.1 Define and describe fundamental pharmacologic concepts, including pharmacokinetics, pharmacodynamics, drug-receptor interactions, dose-response relationships, therapeutic index, half-life, and steady state.

Part 3 · PK Calculations: Half-Life, Steady State & Clearance

IO1.2 Describe the processes of drug absorption, distribution, metabolism, and excretion and factors that influence these processes.

Part 1 · Drug Metabolism & CYP450 Interactions
Part 2 · Drug Excretion & Renal Dosing

IO1.3 Explain how patient-specific factors, including age, pregnancy, renal and hepatic function, and pharmacogenetic variation, may affect medication response.

⚠ Age and pregnancy are only named in the prescribing checklist

Part 2 · Drug Excretion & Renal Dosing
Part 3 · PK Calculations: Half-Life, Steady State & Clearance
Part 4 · Pharmacogenomics
Part 5 · Putting It All Together: Clinical Decision-Making

IO1.4 Identify common types and mechanisms of adverse drug reactions, drug-drug interactions, and drug-disease interactions.

Part 1 · Drug Metabolism & CYP450 Interactions
Part 4 · Pharmacogenomics
Part 5 · Putting It All Together: Clinical Decision-Making

IO1.5 Interpret common components of medication orders and prescriptions, including drug name, dose, route, frequency, and duration.

⚠ Only dose, route, interval, and monitoring (checklist, renal dosing); no prescription-reading content

Part 2 · Drug Excretion & Renal Dosing
Part 5 · Putting It All Together: Clinical Decision-Making

SO1 Explain drug biotransformation (Phase I/II metabolism) and the clinical impact of CYP450 enzyme inhibition and induction.

Part 1 · Drug Metabolism & CYP450 Interactions

SO2 Explain mechanisms of drug excretion and how renal function affects dosing.

Part 2 · Drug Excretion & Renal Dosing

SO3 Calculate and apply basic pharmacokinetic parameters (half-life, steady state, clearance).

Part 3 · PK Calculations: Half-Life, Steady State & Clearance

SO4 Describe how pharmacogenomic variation can alter drug response.

Part 4 · Pharmacogenomics

SO5 Apply a structured framework to clinical prescribing decisions.

Part 5 · Putting It All Together: Clinical Decision-Making

ABOUT THIS LECTURE

- Session 2 of 2 on Introduction to Pharmacology (Thursday, October 22): metabolism, excretion, PK calculations, pharmacogenomics & clinical integration. S1

- Recap: Session 1 covered pharmacology vocabulary, drug development, pharmacodynamics, dose-response relationships, and ADME through distribution. Today picks up with metabolism (Section 6) and runs through the integrated clinical framework and recall quiz. S2; S2 (notes)
- The Lecture Recall Quiz is a cumulative self-test covering both sessions — a standalone recall resource, separate from the in-lecture quick checks; revisit it on your own before the exam. (Rebuilt here as practice questions.) S45; S48 (notes)
- **Original figures in this deck:** Figures 6–8 (CYP450 inhibition vs. induction; exponential decay over half-lives; reaching steady state with and without a loading dose) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts — original diagrams, not reproduced from Katzung or any other copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S49; S49 (notes)

SLIDE DISCREPANCIES

- **Time to steady state:** slides say ~4–5 half-lives; the textbook's rule of thumb is four half-lives (over 90% of steady state). These agree — the textbook just rounds down. S23; p. 47

1 PART 1 · TOPIC

Drug Metabolism & CYP450 Interactions

SO1 IO1.2 IO1.4

Key definitions

- Drug metabolism = how the body chemically transforms drugs. S3
- Phase I — Functionalization: oxidation, reduction, hydrolysis. S5
- Phase II — Conjugation: glucuronidation, sulfation, acetylation, methylation, glutathione conjugation. S5
- **Enzyme inhibition:** a drug slows a CYP enzyme, raising the blood level of any drug that CYP normally clears. S6; SG U1-R10
- **Enzyme induction:** a drug increases CYP enzyme production, lowering the blood level (and effect) of drugs that CYP clears. S6; SG U1-R10
- TB **Microsomal mixed-function oxidase system:** the Phase I machinery in the smooth endoplasmic reticulum; needs cytochrome P450, NADPH-P450 reductase, NADPH and molecular oxygen. p. 59
- TB **Cytochrome P450** is named for the 450-nm light absorption of its carbon-monoxide complex. It has very low substrate specificity: high lipid solubility is the one thing its substrates share. p. 59; p. 60

How it works

- Why metabolize? To inactivate an active drug, making it easier to excrete. S4
- To activate a prodrug into its active form. S4
- To produce an active metabolite that contributes to (or extends) the drug effect. S4
- **Occasionally to produce a toxic metabolite — the basis of some drug-induced organ injury.** S4
- Phase I is primarily carried out by cytochrome P450 (CYP450) enzymes and often creates or unmasks a reactive functional group. S5
- Phase II attaches a polar group to the molecule, producing a more water-soluble product for renal/biliary excretion. S5
- Phase I often comes before Phase II, but not always — some drugs go straight to conjugation. S5 (notes)
- Inhibition slows clearance (substrate level rises); induction speeds clearance (substrate level falls). S6; SG U1-R10
- TB CYP3A4 alone metabolizes over 50% of the prescription drugs metabolized by the liver. p. 60
- TB Acetaminophen is mostly conjugated (glucuronide/sulfate); in overdose glutathione is depleted and the toxic metabolite (NAPQI) injures the liver — **N-acetylcysteine** given within 8–16 hours protects against fulminant hepatotoxicity. p. 66
- TB Share of liver P450: CYP3A4 about 30%, CYP2C9 about 20%, CYP1A2 about 15%, CYP2E1 about 10%, CYP2D6 about

5%. CYP3A4 handles over 50% of liver-metabolized prescription drugs. p. 60

- TB Inducers act through receptors that switch on CYP genes: **AhR** (tobacco smoke, charcoal-broiled meat, cruciferous vegetables, omeprazole → CYP1A) and **PXR** (rifampin, St. John's wort, dexamethasone, phenobarbital → CYP3A). p. 60
- TB **Commensal gut microbiota** metabolize drugs too: they reduce digoxin, so antibiotics such as erythromycin or tetracycline can double digoxin levels; bacterial β-glucuronidases free glucuronidated drugs for reabsorption (enterohepatic recycling). p. 71
- TB An inducer can speed up its own metabolism, producing **pharmacokinetic tolerance**: the same dose works less over time. p. 72

Drugs named as examples

- NAPQI (from acetaminophen) is the textbook example of a toxic metabolite produced by normal Phase I metabolism — previewing the acetaminophen toxicity lecture. S4 (notes)
- CYP3A4 — the enzyme in the lecture's inhibitor/substrate example. S6
- TB **Inducers:** rifampin, carbamazepine, phenytoin, barbiturates, St. John's wort; **inhibitors:** ketoconazole, clarithromycin/erythromycin, ritonavir, grapefruit juice. p. 63; p. 64
- TB St. John's wort induces CYP3A4 and speeds metabolism of oral contraceptives, warfarin, cyclosporine, and others. p. 72
- TB CYP3A4 inhibitors (ketoconazole, erythromycin, grapefruit juice) with terfenadine caused fatal torsades de pointes, and terfenadine was withdrawn. p. 72
- TB Allopurinol inhibits xanthine oxidase, so the mercaptopurine dose must be reduced when both are given. p. 72

CLINICAL

- CYP450 interactions are one of the highest-yield concepts in all of pharmacology — nearly every major drug class has at least one clinically important CYP interaction. S6 (notes)
- **Example: Drug A inhibits CYP3A4 → Drug B (a CYP3A4 substrate) accumulates → toxicity risk rises.** S6; SG U1-R10
- Inhibition pushes a substrate's level ABOVE 100% of baseline (toxicity risk); induction pushes it BELOW 100% (therapeutic-failure risk). S7 (notes)
- Inhibiting the CYP that clears a substrate is especially dangerous when the substrate has a narrow therapeutic index. S11
- Enzyme induction speeds substrate clearance, lowering its level and effect — the opposite of inhibition. S48
- TB Liver disease (e.g., cirrhosis, hepatitis) slows metabolism of some drugs enough that ordinary doses can cause coma. p. 73
- TB Severe ADRs contribute to about 100,000 US deaths a year and about 7% of hospital admissions. p. 71
- TB Smoking induces CYP1A enzymes and can change the metabolism of other drugs. p. 60
- TB **Diet and environmental factors** change drug metabolism: charcoal-broiled food and cruciferous vegetables induce CYP1A, grapefruit juice inhibits CYP3A, smokers and pesticide-exposed workers clear some drugs faster, which complicates dosing of narrow-TI drugs. p. 71
- TB Stopping an inducer is as risky as starting one: when a sedative inducer is stopped, warfarin metabolism slows, levels rise and bleeding can follow. p. 72
- TB Pharmacogenomic testing is still rarely used, although severe ADRs cause about 100,000 US deaths a year and about 7% of hospital admissions. p. 71

TRAP

- Inhibitor → substrate level UP (toxicity); inducer → substrate level DOWN (therapeutic failure). Don't swap them. S7 (notes); S9
- CYP3A4 is a Phase I (CYP450) enzyme — it does not "only affect Phase II metabolism." S5; S10-11
- Not every drug goes through Phase I first — some go straight to Phase II conjugation. S5 (notes)

Step by step

Two phases of metabolism S5

Drug → Phase I — functionalization (CYP450) →

Phase II — conjugation (adds polar group) → More water-soluble product →

Renal/biliary excretion

CYP inhibition S6; S8

Drug A inhibits CYP3A4 → Drug B (CYP3A4 substrate) cleared more slowly →

Drug B accumulates (Figure 6: 240%) → Toxicity risk ↑

CYP induction S6; S7 (notes); S8

Inducer ↑ CYP enzyme production → Substrate cleared faster →

Level and effect ↓ (Figure 6: 42%) → Therapeutic-failure risk ↑

TB Acetaminophen: normal dose vs. overdose (textbook) p. 66

Acetaminophen → Mostly glucuronide/sulfate conjugates → excreted →

Small fraction → CYP → NAPQI, detoxified by glutathione →

Overdose: glutathione depleted → NAPQI injures the liver →

N-acetylcysteine within 8–16 h protects

Figure 6 · CYP450 inhibition vs. induction. Original illustrative bar chart (illustrative values): a CYP450 inhibitor raises a substrate drug's blood level (to 240% of baseline — toxicity risk); an inducer lowers it (to 42% — therapeutic-failure risk). S7; S7 (notes); S8

Phase I vs. Phase II metabolism

	Phase I — Functionalization	Phase II — Conjugation
Reactions	Oxidation, reduction, hydrolysis	Glucuronidation, sulfation, acetylation, methylation, glutathione conjugation
Main enzymes	Cytochrome P450 (CYP450)	Not covered in materials
What it does	Creates or unmasks a reactive functional group	Attaches a polar group
Result	Not covered in materials	More water-soluble product for renal/biliary excretion
Order	Often first — but not always	Some drugs go straight here

How to tell them apart. Sources: S5; S5 (notes)

CYP450 inhibition vs. induction

	Enzyme inhibition	Enzyme induction
What happens to the CYP enzyme	Slowed	More enzyme produced
Substrate drug's blood level	**Rises** (Figure 6: 240%)	**Falls** (Figure 6: 42%)

	Enzyme inhibition	Enzyme induction
Clinical risk	**Toxicity**	**Therapeutic failure**
Worst case	Narrow-TI substrate	Not covered in materials

How to tell them apart. Sources: S6; S7 (notes); S8-9; S11

How inducers switch on CYP enzymes (textbook Ch. 4)

Receptor	Inducers	Enzyme induced
AhR	Tobacco smoke, charcoal-broiled meat, cruciferous vegetables, omeprazole	CYP1A
PXR	Rifampin, St. John's wort (hyperforin), dexamethasone, phenobarbital	CYP3A
CAR	Phenobarbital-class inducers	CYP2B6, CYP2C9

TB How to tell them apart. Sources: p. 60

WHY IT HAPPENS

- Why does an inhibitor cause toxicity? The CYP enzyme that normally clears the substrate is slowed, so the substrate can no longer be cleared normally and accumulates. S6; S9
- Why does Phase II help excretion? Attaching a polar group makes the drug more water-soluble, so it can leave in urine or bile. S5

MNEMONIC (MEMORY AID)

- **IN**hibitor = level goes **IN**to the danger zone (↑, toxicity). In**DU**cer = level goes **D**own (↓, failure). S6; S7 (notes)
- Phase **I** = **I**nitial change (oxidize, reduce, hydrolyze — CYP450). Phase **II** = **T**ag it (conjugate a polar group → water-soluble). S5

HIGH-YIELD

- Phase I (CYP450): oxidation, reduction, hydrolysis. Phase II: conjugation → water-soluble. S5
- **Inhibitor → substrate ↑ → toxicity**; **inducer → substrate ↓ → therapeutic failure**. S6; S7 (notes)
- Narrow-TI substrate + CYP inhibitor = especially dangerous. S11
- Metabolism can produce toxic metabolites (e.g., NAPQI). S4; S4 (notes)

2 PART 2 · TOPIC

Drug Excretion & Renal Dosing

SO2 IO1.2 IO1.3 IO1.5

Key definitions

- Drug excretion = clearing drug from the body. S12
- **Glomerular filtration:** free (unbound) drug passes from blood into the renal tubule based on molecular size. S14
- **Tubular secretion:** active transporters move additional drug from blood into the tubule. S14
- **Tubular reabsorption:** lipophilic, un-ionized drug can passively diffuse back into the blood, reducing net elimination. S14

How it works

- Kidney: the dominant route for most drugs and metabolites. S13
- Bile / feces: important for some large or protein-bound drugs. S13
- Lungs: volatile agents (e.g., anesthetic gases). S13
- Sweat / breast milk: minor routes, but breast milk matters for lactating patients. S13
- **Decreased renal function → decreased drug clearance → increased drug exposure and accumulation risk.** S15

Drugs named as examples

- Renally-cleared, narrow-TI drugs: many antibiotics, digoxin, lithium. S15
- Aspirin overdose — urine alkalinization traps ionized drug. S14 (notes)
- Anesthetic gases — eliminated by the lungs. S13

CLINICAL

- eGFR and creatinine clearance (CrCl) are the standard bedside estimates of renal function used to adjust dosing. S15
- Any renally-cleared, narrow-therapeutic-index drug (think: many antibiotics, digoxin, lithium) needs a dose or interval adjustment in renal impairment. S15
- For a primarily renally eliminated drug when eGFR drops: decrease the dose and/or extend the dosing interval, guided by eGFR/CrCl. S16-17
- Urine pH manipulation (e.g., alkalinizing urine in aspirin overdose) works by trapping ionized drug in the tubule, blocking reabsorption. S14 (notes)
- This is the PK foundation for the renal-impairment case and the NSAID-in-CKD case later. S15 (notes)

TRAP

- Only FREE (unbound) drug is filtered at the glomerulus. S14
- Reabsorption REDUCES net elimination — and only lipophilic, un-ionized drug is reabsorbed. S14
- Renal impairment → lower dose and/or longer interval — NOT "no adjustment," a higher dose, or "IV only." S16-17

Step by step

Renal handling of a drug S14

Free drug filtered at glomerulus (by size) →

Transporters secrete more into tubule → Lipophilic, un-ionized drug reabsorbed

→ Remainder eliminated in urine

Three renal processes

Process	Direction	How	Which drug
Glomerular filtration	Blood → tubule	Based on molecular size	Free (unbound) drug
Tubular secretion	Blood → tubule	Active transporters	Additional drug
Tubular reabsorption	Tubule → blood	Passive diffusion — **reduces net elimination**	Lipophilic, un-ionized drug

How to tell them apart. Sources: S14

Routes of drug elimination

Route	Importance	Example / who it matters for
Kidney	**Dominant** for most drugs and metabolites	Not covered in materials
Bile / feces	Important for some	Large or protein-bound drugs
Lungs	Specific agents	Volatile agents (anesthetic gases)
Sweat / breast milk	Minor	Breast milk matters for lactating patients

How to tell them apart. Sources: S13

WHY IT HAPPENS

- Why does alkalinizing urine help in aspirin overdose? It keeps the drug ionized in the tubule, and only un-ionized drug can diffuse back — so reabsorption is blocked and more drug leaves. S14; S14 (notes)
- Why adjust doses in CKD? Less renal function → less clearance → more exposure and accumulation of renally-cleared drugs. S15

MNEMONIC (MEMORY AID)

- Kidney handling = "**F**ilter, **S**ecrete, **R**eabsorb" — **F**ree drug filtered, **S**ecreted by transporters, **R**eabsorbed if lipophilic & un-ionized. S14

HIGH-YIELD

- Kidney = dominant elimination route; filtration (free drug), secretion (transporters), reabsorption (lipophilic, un-ionized). S13; S14
- **↓ renal function → ↓ clearance → ↑ exposure** → lower dose and/or longer interval, guided by **eGFR/CrCl**. S15; S17
- Renally cleared narrow-TI drugs: many antibiotics, digoxin, lithium. S15

3 PART 3 · TOPIC

PK Calculations: Half-Life, Steady State & Clearance

SO3 IO1.1 IO1.3

Key definitions

- **Half-life (t½) = the time required for the plasma drug concentration to fall by 50%.** S19
- **Steady state:** the point at which drug intake equals drug elimination; reached in about 4–5 half-lives with regular dosing. S23; SG U1-R11
- **Loading dose:** a larger initial dose used to rapidly reach a therapeutic concentration, without waiting for steady state. S23
- **Maintenance dose:** the regular, ongoing dose that keeps the drug within the therapeutic range once steady state is reached. S23
- **Clearance = the volume of plasma from which drug is completely removed per unit time.** S27
- TB **t½ = 0.7 × Vd / CL** — half-life rises when Vd rises or clearance falls. p. 56
- TB **Loading dose = Vd × target concentration** (e.g., theophylline: 35 L × 10 mg/L = 350 mg for a 70-kg adult). p. 52
- TB **Extraction ratio (ER):** hepatic clearance divided by hepatic blood flow (about 90 L/h). Oral bioavailability ≈ fraction absorbed × (1 − ER); morphine's ER of 0.67 predicts about 33% oral bioavailability. p. 49
- TB **Target concentration approach** (the rational way to design a dosage regimen): pick the concentration that gives the desired effect, then individualize the dose from the patient's clearance and volume of distribution. p. 51

How it works

- PK calculations = the math PAs actually use at the bedside. S18
- Half-life is determined by clearance and volume of distribution together — it is not a fixed drug "property" independent of physiology. S19
- A drug is roughly 97% eliminated after about 5 half-lives. S19; SG U1-R11
- **Formula: % remaining = 100% x (1/2)^(number of half-life)** — i.e., per number of half-lives elapsed. S19
- **Formula: $C(t) = C_o \times (1/2)^{(t/t½)}$** — C_o = starting concentration, t = time elapsed. S19
- Worked example (both formulas): t½ = 6 h; an IV dose gives a starting plasma concentration of 100 mg/L — see the table for each half-life. S19
- Clearance determines how efficiently a drug is removed — and, along with Vd, determines half-life. S27
- Total clearance is usually the sum of hepatic and renal clearance. S27
- Loading doses are based on Vd; maintenance doses are based on clearance. S23 (notes)
- Steady state takes ~4–5 half-lives regardless of dose size — the same reason a drug is ~97% eliminated ~5 half-lives after stopping it. S30
- Time to steady state (about 4–5 half-lives) is governed by half-life. S48
- TB During regular dosing: 50% of steady state after 1 half-life, 75% after 2, and over

- TB **Zero-order absorption** runs at a constant rate (gastric emptying, controlled-release forms); **first-order** absorption is proportional to the amount left, nearly complete after four absorption half-lives. p. 49

90% after 4 — the textbook's "four half-lives" rule of thumb. p. 47

- TB If the dosing interval is shorter than four half-lives, accumulation will be detectable. p. 48
- TB **Capacity-limited (saturable) elimination:** clearance falls as concentration rises — important for phenytoin, ethanol, and aspirin. p. 47
- TB **Maintenance dosing rate = clearance × target concentration** (divide by F for oral). The average steady-state level depends only on clearance; the dosing interval changes only the peaks and troughs. p. 51
- TB Effects do not always follow plasma levels: enalapril (half-life about 4 hours) still works once daily because its peak is far above its C_{50}; delays come from distribution, slow receptor release (digoxin) or slow turnover of a body substance (warfarin and clotting factors). p. 50
- TB Half-life can change without clearance changing: diazepam's half-life lengthens with age because its volume of distribution grows, not because metabolism slows. p. 54

Drugs named as examples

- Not covered in materials — this section uses illustrative drugs ("a drug with a half-life of 6 hours"). S19
- TB Theophylline worked example: clearance 2.8 L/h × target 10 mg/L = 28 mg/h IV; loading dose = 35 L × 10 mg/L = 350 mg. p. 52

CLINICAL

- Case: significant CKD + a medication primarily eliminated unchanged by the kidney. **What happens to the drug's concentration and half-life, and would you adjust the dose, the dosing interval, or both?** S28
- **Clinical pearl:** reduced clearance raises drug concentration and prolongs half-life; many renally-cleared drugs are managed by extending the dosing interval, reducing the dose, or both, guided by eGFR/CrCl. S28
- The logic chain: clearance ↓ → half-life ↑ → time to steady state ↑ → accumulation risk ↑ if the regimen is not adjusted. S28 (notes)

- Reduced renal clearance raises both drug concentration and half-life. S48
- A loading dose reaches the therapeutic target immediately instead of waiting ~4–5 half-lives; it does not change the maintenance dose or remove the need for monitoring. S26
- TB Highly extracted drugs (morphine, isoniazid, propranolol, lidocaine, verapamil) vary widely in bioavailability between patients, and much more drug gets through in cirrhosis with portosystemic shunting. Lidocaine is never given orally because of toxic metabolites. p. 49
- TB **Schedule-dependent effects:** the same average level can carry different toxicity; gentamicin is more nephrotoxic as a constant infusion than as intermittent doses, because high peaks saturate its uptake into the renal cortex. p. 50
- TB Give IV loading doses slowly (over an hour, not seconds): a fast bolus can overshoot into toxicity before the drug distributes, e.g., phenytoin. p. 52
- TB Timing drug levels: digoxin at least 6 hours after the dose, lithium just before the next dose; for steady-state estimates sample mid-interval after at least three half-lives. p. 55
- TB In obesity, dose drugs that stay out of fat (gentamicin, digoxin) on fat-free mass; edema and ascites enlarge gentamicin's volume of distribution. p. 53; p. 55
- TB Changes in plasma protein binding rarely matter clinically: the body simply eliminates the extra unbound drug. p. 55

TRAP

- After 3 half-lives, 12.5% REMAINS (87.5% eliminated) — read whether the question asks remaining or eliminated. S19; S22
- Time to steady state depends on half-life — not on dose size, bioavailability, or route. S30; S48
- Half-life is not a fixed property — it changes with clearance and Vd (e.g., in renal impairment). S19; S28
- A loading dose does NOT permanently lower the maintenance dose or eliminate monitoring. S26

Step by step

Renal impairment logic chain S28 (notes)

Clearance ↓ → Half-life ↑ → Time to steady state ↑ →

Accumulation risk ↑ (if regimen not adjusted)

Half-life decay (100 mg/L, t½ 6 h) S19

0 h: 100 mg/L → 6 h: 50 → 12 h: 25 → 18 h: 12.5 → 24 h: 6.25 →

30 h: 3.125 (~97% gone)

Figure 7 · Exponential decay over half-lives. Original illustrative decay curve: plasma concentration falls by 50% with each half-life, leaving ~3% remaining after 5 half-lives. S20

Figure 8 · Steady state with and without a loading dose. Original illustrative multi-dose simulation: maintenance dosing alone climbs gradually (a sawtooth climb) to steady state; adding a loading dose reaches the target level immediately, right from the first dose. S24; S24 (notes)

TB **Textbook concept · Five to fill, five to empty.** Original diagram: starting regular dosing, each half-life closes half of the remaining gap to steady state (50 → 75 → 87.5 → 93.75 → ~97%); after stopping, each half-life removes half of what remains (50 → 25 → 12.5 → 6.25 → 3.125%). The same 4–5 half-lives govern both. S19; S23; p. 47

Half-life worked example (t½ = 6 h, start 100 mg/L)

Half-lives	Time	Concentration	% remaining	% eliminated
1	6 h	50 mg/L	50%	50%
2	12 h	25 mg/L	25%	75%
3	18 h	12.5 mg/L	12.5%	87.5%
4	24 h	6.25 mg/L	6.25%	93.75%
5	30 h	3.125 mg/L	3.125%	**~97%**

How to tell them apart. Sources: S19

Loading dose vs. maintenance dose

	Loading dose	Maintenance dose
Purpose	Reach therapeutic concentration rapidly	Keep the drug in the therapeutic range
Size / timing	Larger, initial	Regular, ongoing
Based on	**Vd**	**Clearance**

	Loading dose	Maintenance dose
On Figure 8	Starts at target immediately	Sawtooth climb over ~4–5 half-lives

How to tell them apart. Sources: S23; S23 (notes); S24 (notes)

Dosing from the target concentration (textbook Ch. 3)

Goal	Formula	Depends on
Keep the level steady	Maintenance dosing rate = CL × target concentration (÷ F if oral)	Clearance
Reach the target now	Loading dose = Vd × target concentration	Volume of distribution
Match the peak at steady state	Loading dose = maintenance dose × accumulation factor	Dosing interval and half-life

TB How to tell them apart. Sources: p. 51; p. 52

WHY IT HAPPENS

- Why does CKD prolong half-life? Half-life depends on clearance (with Vd) — less renal clearance means the concentration takes longer to fall by 50%, so the drug accumulates. S19; S27; S28 (notes)
- Why does a loading dose skip the wait? It is a larger first dose that reaches the therapeutic concentration right away instead of building up over ~4–5 half-lives. S23; S26

MNEMONIC (MEMORY AID)

- "**Five to fill, five to empty**": ~4–5 half-lives to reach steady state; ~5 half-lives to be ~97% eliminated. S19; S23; S30
- **L**oading → **L**iters (Vd). **M**aintenance → **M**oving out (clearance). S23 (notes)

HIGH-YIELD

- **t½** = time for plasma concentration to fall 50%; set by **clearance + Vd**. S19
- Each t½ halves what remains: 50 → 25 → 12.5 → 6.25 → ~3% (~97% eliminated at 5 t½). S19; S22
- **Steady state ≈ 4–5 t½**, regardless of dose; loading dose gets there immediately. S23; S26; S30
- **Clearance** = plasma volume cleared per unit time; total = hepatic + renal. S27
- CKD: clearance ↓ → t½ ↑ → concentration ↑ → adjust dose and/or interval. S28; S28 (notes)

4 PART 4 · TOPIC

Pharmacogenomics

SO4 IO1.3 IO1.4

Key definitions

- **Pharmacogenomics studies how genetic variation changes an individual's response to a drug.** S32
- Why the same dose does not work the same way in everyone. S31
- TB **Allele:** one of the alternative forms of a gene at the same locus. **Haplotype:** a series of alleles inherited together on one chromosome. **Nonsynonymous SNP:** a single-base change that alters an amino acid. p. 78
- TB **PM, IM, EM, UM:** poor, intermediate, extensive (normal) and ultrarapid metabolizer. p. 78

How it works

- Genetic variation can alter drug metabolism (e.g., CYP enzyme activity). S32
- It can alter drug transport across membranes. S32
- It can alter drug target structure or expression. S32
- It can alter overall drug response and risk of adverse reactions. S32
- Personalized pharmacotherapy combines genotype, phenotype, disease state, and other medications into individualized therapy. S34
- **Pharmacogenomics is one input among several — genotype alone never dictates therapy in isolation.** S34
- TB Codeine is converted to morphine (200× more potent) by CYP2D6; CPIC recommends an alternative analgesic for CYP2D6 poor **and** ultrarapid metabolizers. p. 81; p. 83
- TB Clopidogrel: about 85% of a dose is inactivated by esterases; CYP2C19 activates the rest. CYP2C19 poor/intermediate metabolizers → prasugrel or ticagrelor. p. 81; p. 82
- TB TPMT intermediate activity: start thiopurines at 30–70% of the target dose; deficient: use a drastically reduced dose or an alternative. p. 83
- TB Warfarin dose varies up to 20-fold between patients; CYP2C9 and VKORC1 genotypes explain much of it. p. 90
- TB **G6PD deficiency** (X-linked, more than 400 million people): G6PD is the red cell's only source of NADPH, so deficient

cells hemolyze under oxidative stress such as infection, fava beans or certain drugs. p. 86

- TB NUDT15 variants, as well as TPMT, cause thiopurine intolerance. p. 86
- TB **OCT1 (organic cation transporter 1)** moves basic drugs such as morphine, tramadol, sumatriptan and ondansetron into the liver; people with two reduced-function alleles have about twofold higher levels. p. 87

Drugs named as examples

- Codeine (CYP2D6), clopidogrel (CYP2C19), azathioprine (TPMT), allopurinol (HLA-B*5801); also antidepressants, opioids, PPIs. S33; S36
- TB CYP2D6 metabolizes up to one-quarter of clinically used drugs (β blockers, antidepressants, antipsychotics, opioids). p. 78
- TB CYP2D6 ultrarapid metabolizers: up to one-third of Ethiopians and Saudi Arabians; CYP2C19 poor metabolizers: 3–5% of Caucasians, 18–23% of Japanese. p. 67
- TB **HLA-B*57:01** → abacavir hypersensitivity; **G6PD deficiency** → rasburicase contraindicated; slow NAT2 acetylators (~50% in the USA) → isoniazid toxicity. p. 70; p. 83; p. 88
- TB **SLCO1B1 (OATP1B1)** reduced-function variants raise simvastatin exposure (221% higher AUC in homozygotes) and myopathy risk; CPIC advises a lower dose or another statin. p. 87
- TB **ABCG2 (BCRP)** variant, common in East Asians (allele frequency about 30%), raises rosuvastatin levels; the label suggests half the dose. p. 87

CLINICAL

- CYP2D6 variation alters metabolism of many antidepressants and opioids (e.g., codeine activation). S33; SG U1-R12
- CYP2C19 variation affects clopidogrel activation and some antidepressant/PPI dosing. S33

- TPMT deficiency raises toxicity risk with thiopurine drugs (e.g., azathioprine). S33; SG U1-R12
- Certain HLA alleles sharply raise the risk of severe hypersensitivity to specific drugs. S33
- HLA-B*5801 is associated with severe hypersensitivity reactions to allopurinol — a textbook pharmacogenomic screening consideration (seen again in the gout lecture). S36; S33 (notes); SG U1-R12

TRAP

- HLA-B*5801 before allopurinol = severe hypersensitivity risk — not absorption, not taste, and not irrelevant to drug selection. S35-36
- Genotype is one input — never the sole basis for therapy. S34

Step by step

TB Codeine and CYP2D6 (textbook) p. 81; p. 82

Codeine (prodrug) → CYP2D6 → morphine (200× more potent) →

Poor metabolizer: little morphine → poor pain relief →

Ultrarapid: too much morphine → toxicity →

CPIC: alternative analgesic for PM and UM

TB Clopidogrel and CYP2C19 (textbook) p. 81; p. 82

Clopidogrel (prodrug) → ~85% inactivated by esterases →

CYP2C19 makes the active metabolite →

Poor/intermediate metabolizer: less active drug → more stent thrombosis after PCI

→ CPIC: prasugrel or ticagrelor instead

Pharmacogenomics in practice

Variant	What it changes	Drugs named
CYP2D6 variation	Metabolism (incl. codeine **activation**)	Many antidepressants, opioids (codeine)
CYP2C19 variation	Activation / dosing	Clopidogrel; some antidepressants, PPIs

Variant	What it changes	Drugs named
TPMT deficiency	↑ Toxicity risk	Thiopurines (azathioprine)
HLA alleles (HLA-B*5801)	↑↑ Severe hypersensitivity risk	Allopurinol

How to tell them apart. Sources: S33; S36

WHY IT HAPPENS

- Why can a CYP2D6 variant change codeine's effect? Codeine needs CYP2D6 activation — change the enzyme's activity and you change how much active drug is made. S32; S33

MNEMONIC (MEMORY AID)

- Gene → drug pairs: **2D6** → co**D**eine; **2C19** → **C**lopidogrel; **T**PMT → **T**hiopurines; **HLA-B*58** → **A**llopurinol (**A**llergy-like hypersensitivity). S33; S36

HIGH-YIELD

- Pharmacogenomics = genetic variation changing drug response (metabolism, transport, target, ADR risk). S32
- **HLA-B*5801 + allopurinol** → severe hypersensitivity risk. S36
- CYP2D6–codeine; CYP2C19–clopidogrel; TPMT–azathioprine. S33
- Genotype is **one input among several**. S34

5 PART 5 · TOPIC

Putting It All Together: Clinical Decision-Making

SO5 IO1.3 IO1.4 IO1.5

Key definitions

- From concepts to clinical decisions: one integrated pharmacology pathway — Dose → Absorption → Distribution → Receptor / Target → Effect, with Metabolism and Excretion clearing the drug. S37; S38
- **Every topic in this lecture is a stop somewhere along this pathway — use it as your mental checklist when you evaluate any drug.** S38

How it works

- Before you prescribe — the clinical decision-making checklist: Indication — is there a clear reason to treat? S39
- Contraindications and allergies. S39
- Renal and hepatic function. S39
- Drug interactions (including CYP450). S39
- Age, pregnancy, and other patient-specific factors. S39
- Correct dose, route, and monitoring plan. S39
- Memorize this list — it is exactly the checklist applied in the MSK cases (NSAID in CKD, methotrexate in pregnancy, etc.). S39 (notes)

Drugs named as examples

- Not covered in materials — the integrated vignettes use unnamed drugs ("a CYP3A4-substrate medication," "a home medication"). S40; S41

CLINICAL

- CYP interaction in action: a patient stable on a CYP3A4-substrate medication starts a potent CYP3A4 inhibitor. **What happens to the first drug's level, and what would you watch for clinically?** S40
- **Clinical pearl: always check for CYP450 interactions before adding a new drug to an existing regimen — one of the most preventable causes of toxicity.** S40; S43
- Agonist or antagonist? A new drug binds the same receptor as a home medication, and the home medication's effect weakens. **Is the new drug most likely a competitive antagonist, and how could you confirm it pharmacologically?** S41

- **Clinical pearl: a competitive antagonist can be overcome by increasing the agonist dose — if raising the home medication's dose restores the effect, that supports competitive (rather than irreversible) antagonism.** S41
- SG Study-guide case: 68-year-old woman, stage 4 CKD (eGFR 24 mL/min), starting a narrow-TI oral antiarrhythmic eliminated mainly unchanged by the kidney; she started a strong CYP3A4 inhibitor 3 weeks ago. SG U1-S
- SG Two concerns: reduced renal clearance from CKD will raise the antiarrhythmic's concentration and prolong its half-life regardless of the new drug; separately, if it has any hepatic/CYP3A4 component, the CYP3A4 inhibitor would compound accumulation further. SG U1-S
- SG Before dispensing: check eGFR/CrCl, consider a lower dose or extended interval, and closely monitor renal function and signs of toxicity (arrhythmia worsening, other drug-specific toxicity) after starting. SG U1-S
- SG If the drug were instead cleared by the liver via CYP3A4: the direct renal-clearance concern becomes less central (though CKD can still indirectly affect hepatic drug handling), and the CYP3A4 interaction becomes the dominant concern. SG U1-S

TRAP

- The single check most aimed at preventing a toxic, predictable drug-drug interaction is reviewing CYP450 interactions — not brand name, chemical name, or half-life alone. S42-43
- Weaker home-medication effect + restored by a higher dose = competitive; not restored = irreversible. S41
- Reviewing indication, contraindications, renal/hepatic function, interactions, and monitoring = the clinical decision-making framework — not pharmacogenomic testing or Phase IV surveillance. S47-48

Step by step

One integrated pathway S38

Dose → Absorption → Distribution → Receptor / target → Effect → Metabolism → Excretion

Key takeaways: the toolkit for every drug class

#	Idea
1	Pharmacodynamics = what the drug does to the body (receptors, potency, efficacy).
2	Pharmacokinetics = what the body does to the drug (ADME).
3	Receptor interactions (agonism, antagonism) explain most drug effects.

#	Idea
4	ADME together determine how much drug reaches the target, and for how long.
5	Patient-specific factors — renal/hepatic function, genetics, interactions — determine the individual response.

How to tell them apart. Sources: S44; S44 (notes)

WHY IT HAPPENS

- Why check CYP450 before adding any drug? An inhibitor added to a stable regimen raises the other drug's level — predictable, so it is one of the most preventable causes of toxicity. S6; S40

MNEMONIC (MEMORY AID)

- Checklist "**I C R I A D**": **I**ndication · **C**ontraindications/allergies · **R**enal/hepatic · **I**nteractions (CYP450) · **A**ge/pregnancy · **D**ose, route, monitoring. S39

HIGH-YIELD

- Memorize the checklist: indication, contraindications/allergies, renal/hepatic, interactions (CYP450), age/pregnancy, dose/route/monitoring. S39; S39 (notes)
- **Always check CYP450 interactions before adding a new drug.** S40
- These five ideas are the toolkit for every drug class for the rest of the course, starting with musculoskeletal pharmacology. S44 (notes)

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
4 purposes of metabolism	Inactivate drug · activate prodrug · active metabolite · occasionally toxic metabolite S4
NAPQI	Toxic metabolite from acetaminophen via normal Phase I metabolism S4 (notes)
Phase I	Functionalization: oxidation, reduction, hydrolysis; mainly CYP450; unmasks reactive group S5
Phase II	Conjugation: glucuronidation, sulfation, acetylation, methylation, glutathione → water-soluble S5
Phase order	Phase I often first — but some drugs go straight to conjugation S5 (notes)
CYP enzyme inhibition	Slows CYP → substrate level ↑ → toxicity risk S6
CYP enzyme induction	More CYP made → substrate level ↓ → therapeutic failure S6; S7 (notes)
Figure 6 values	Inhibitor: 240% of baseline; inducer: 42% S8
CYP3A4 inhibitor + narrow-TI substrate	Level ↑ and toxicity risk ↑ — especially dangerous S11
Main elimination route	Kidney (most drugs and metabolites) S13
Bile / feces	Some large or protein-bound drugs S13
Lungs	Volatile agents (anesthetic gases) S13
Sweat / breast milk	Minor; breast milk matters in lactation S13
Glomerular filtration	Free (unbound) drug, by molecular size S14
Tubular secretion	Active transporters move more drug into tubule S14
Tubular reabsorption	Lipophilic, un-ionized drug diffuses back → less elimination S14
Alkalinize urine (aspirin overdose)	Traps ionized drug in tubule, blocks reabsorption S14 (notes)
Renal function estimates	eGFR and creatinine clearance (CrCl) S15
Renally cleared narrow-TI drugs	Many antibiotics, digoxin, lithium — adjust dose or interval S15
CKD dosing	Decrease dose and/or extend interval, guided by eGFR/CrCl S17; S28
Half-life (t½)	Time for plasma concentration to fall 50%; set by clearance + Vd S19

Prompt	Answer
% remaining formula	100% x (1/2)^(number of half-lives) S19
Concentration formula	$C(t) = C_o$ x (1/2)^(t/t½) S19
Remaining after 1/2/3/4/5 t½	50% / 25% / 12.5% / 6.25% / 3.125% S19
~97% eliminated	After about 5 half-lives S19; S30
Steady state	Intake = elimination; ~4–5 half-lives, regardless of dose S23; S30
Time to steady state set by	Half-life S48
Loading dose	Larger first dose → target immediately; based on **Vd** S23; S23 (notes); S26
Maintenance dose	Ongoing dose to stay in range; based on **clearance** S23; S23 (notes)
Clearance	Plasma volume completely cleared of drug per unit time; total = hepatic + renal S27
CKD effect on PK	Clearance ↓ → t½ ↑ → concentration ↑ → time to steady state ↑ S28; S28 (notes)
Pharmacogenomics	Genetic variation changing drug response (metabolism, transport, target, ADR risk) S32
CYP2D6 variation	Antidepressants, opioids — codeine activation S33
CYP2C19 variation	Clopidogrel activation; some antidepressant/PPI dosing S33
TPMT deficiency	↑ Toxicity with thiopurines (azathioprine) S33
HLA-B*5801	Severe hypersensitivity to allopurinol S36
Personalized therapy inputs	Genotype + phenotype + disease state + other medications S34
Prescribing checklist	Indication · contraindications/allergies · renal/hepatic · interactions (CYP450) · age/pregnancy · dose/route/monitoring S39
Effect weakened, restored by higher agonist dose	Competitive (not irreversible) antagonism S41
Single best check vs. predictable toxicity	Review CYP450 interactions with current medications S43
t½ formula	t½ = 0.7 × Vd / CL p. 56
Loading dose formula	Vd × target concentration p. 52
% of steady state	1 t½ = 50% · 2 = 75% · 4 = >90% p. 47
Saturable elimination	Phenytoin, ethanol, aspirin p. 47
CYP3A4 share	>50% of liver-metabolized prescription drugs p. 60

Prompt	Answer
Acetaminophen antidote	N-acetylcysteine within 8–16 h p. 66
CYP2C19 PM on clopidogrel	Switch to prasugrel or ticagrelor p. 82
HLA-B*57:01	Abacavir hypersensitivity p. 88
Extraction ratio	Hepatic CL ÷ hepatic blood flow; F ≈ f × (1 − ER) p. 49
Maintenance dosing rate	CL × target concentration (÷ F orally) p. 51
Average steady-state level	Set by clearance alone p. 51
CYP450 machinery	P450 + reductase + NADPH + O_2 (smooth ER) p. 59
Smoking	Induces CYP1A (AhR) p. 60
Digoxin level timing	At least 6 h after the dose p. 55
SLCO1B1 variant	Simvastatin myopathy: lower dose or another statin p. 87

Self-test

First, the lecture's own 14 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S8-9

In Figure 6, the "+ Enzyme inhibitor" bar rises to 240% of baseline, while the "+ Enzyme inducer" bar falls to 42%. For a patient on a stable dose of the substrate drug, which condition poses the greatest risk of toxicity?

- **A** Normal metabolism
- **B** Enzyme induction
- **C** Neither condition changes risk
- **D** Enzyme inhibition — because it raises the substrate drug's blood level

ANSWER: D

The tallest bar in the chart is the enzyme-inhibitor condition at 240% of baseline — the substrate drug accumulates because it can no longer be cleared normally, raising toxicity risk. The inducer bar shows the opposite problem (42%, risking therapeutic failure), not toxicity. S8-9

Lecture check 2 · S10-11

A patient stable on a CYP3A4-substrate medication with a narrow therapeutic index is started on a new drug that is a strong CYP3A4 inhibitor. What is the most likely consequence?

- **A** No change, since CYP3A4 only affects Phase II metabolism
- **B** Decreased substrate drug level and reduced effect
- **C** Increased substrate drug level and increased toxicity risk
- **D** Increased renal clearance of the substrate drug

ANSWER: C

Inhibiting the CYP enzyme responsible for clearing the substrate drug slows its metabolism, raising its blood level — especially dangerous when the substrate has a narrow therapeutic index. S10-11

Lecture check 3 · S16-17

A patient's eGFR drops significantly due to new chronic kidney disease. For a drug that is primarily renally eliminated, what dosing adjustment is most appropriate?

- **A** No adjustment is needed
- **B** Increase the dose
- **C** Switch to IV administration only
- **D** Decrease the dose and/or extend the dosing interval

ANSWER: D

Reduced renal clearance raises drug exposure for renally-eliminated drugs; the standard response is a lower dose, a longer interval, or both, guided by eGFR/CrCl. S16-17

Lecture check 4 · S21-22

Using Figure 7, approximately what percentage of the original drug concentration remains after 3 half-lives (3 t½)?

- **A** 50%
- **B** 25%
- **C** 12.5%
- **D** 3%

ANSWER: C

Each half-life cuts the remaining concentration in half: 100% → 50% (1 t½) → 25% (2 t½) → 12.5% (3 t½), exactly as marked on the curve. By 5 half-lives the drug is roughly 97% eliminated (about 3% remaining). S21-22

Lecture check 5 · S25-26

In Figure 8, the dashed orange line (loading dose + maintenance) starts at the target concentration immediately, while the solid teal line (maintenance only) climbs gradually over several doses. What is the clinical advantage of adding a loading dose?

- **A** It permanently lowers the maintenance dose needed
- **B** It eliminates the need for monitoring
- **C** It has no advantage over maintenance dosing alone
- **D** It reaches the therapeutic target immediately instead of waiting ~4–5 half-lives

ANSWER: D

The dashed loading-dose curve starts at the dotted steady-state line from the very first dose, while the solid maintenance-only curve needs several dosing intervals (~4–5 half-lives) to climb to that same level — exactly as defined on the previous slide. A loading dose does not change the maintenance dose itself or remove the need for monitoring. S25-26

Lecture check 6 · S29-30

Approximately how many half-lives are required for a drug to reach steady state with regular, repeated dosing?

- **A** 1–2
- **B** 4–5
- **C** 8–10
- **D** Steady state is reached immediately after the first dose

ANSWER: B

Steady state — where drug intake equals drug elimination — is reached in roughly 4–5 half-lives, regardless of dose size. This is also why a drug is roughly 97% eliminated about 5 half-lives after stopping it. S29-30

Lecture check 7 · S35-36

A patient is found to carry the HLA-B*5801 allele before starting allopurinol. Why does this matter clinically?

- **A** It identifies a sharply increased risk of a severe hypersensitivity reaction
- **B** It predicts reduced drug absorption
- **C** It only affects how the drug tastes
- **D** It has no bearing on drug selection

ANSWER: A

HLA-B*5801 is associated with severe hypersensitivity reactions to allopurinol — a textbook pharmacogenomic screening consideration you will see again in Thursday's gout lecture. S35-36

Lecture check 8 · S42-43

Before prescribing any new medication, which single check is most directly aimed at preventing a toxic, often-predictable drug-drug interaction?

- **A** Confirming the drug's brand name
- **B** Checking the drug's chemical name
- **C** Reviewing the drug's half-life alone
- **D** Reviewing CYP450 interactions with the patient's current medications

ANSWER: D

CYP450 interaction screening is one of the single highest-yield steps for preventing predictable, preventable toxicity before adding any new drug to a regimen. S42-43

Lecture check 9 · S46; S48

A drug has high potency but low efficacy compared to an alternative. For a patient needing maximum symptom relief, which factor matters more when choosing between the two?

- **A** Efficacy
- **B** Potency
- **C** Both matter equally
- **D** Neither predicts clinical benefit

ANSWER: A

Efficacy (the maximum achievable effect) matters more than potency for most clinical outcomes. S46; S48

Lecture check 10 · S46; S48

A patient with CKD is prescribed a renally-cleared, narrow-TI drug without dose adjustment. Which two consequences would you expect together?

- **A** Decreased half-life and decreased concentration
- **B** Increased clearance and decreased Vd
- **C** Increased half-life and increased concentration
- **D** No change in either parameter

ANSWER: C

Reduced renal clearance raises both drug concentration and half-life. S46; S48

Lecture check 11 · S46; S48

Which best differentiates a competitive antagonist from a noncompetitive antagonist?

- **A** Competitive lowers Emax; noncompetitive does not
- **B** Noncompetitive lowers Emax; competitive shifts the curve right without changing Emax
- **C** They are pharmacologically identical
- **D** Only competitive antagonists cause adverse effects

ANSWER: B

Noncompetitive/irreversible antagonists lower Emax; competitive antagonists shift the curve right without changing Emax. S46; S48

Lecture check 12 · S47; S48

Drug A, a CYP enzyme inducer, is started in a patient taking Drug B, a substrate of that same CYP enzyme. What happens to Drug B's level and effect?

- **A** Increases, raising toxicity risk
- **B** No change
- **C** Drug B's route of elimination changes to biliary
- **D** Decreases, risking therapeutic failure

ANSWER: D

Enzyme induction speeds substrate clearance, lowering its level and effect — the opposite of inhibition. S47; S48

Lecture check 13 · S47; S48

A narrow-therapeutic-index medication requires monitoring. Which pharmacokinetic parameter most directly determines how long it takes to reach steady-state concentration?

- **A** Half-life
- **B** Bioavailability
- **C** Route of administration
- **D** Chemical name

ANSWER: A

Time to steady state (about 4–5 half-lives) is governed by half-life. S47; S48

Lecture check 14 · S47; S48

A PA reviews indication, contraindications, renal/hepatic function, interactions, and a monitoring plan before prescribing. This process best reflects:

- **A** Pharmacogenomic testing
- **B** Receptor pharmacology alone
- **C** The clinical decision-making framework
- **D** Phase IV surveillance

ANSWER: C

This checklist is the clinical decision-making framework from the "Putting It All Together" section. S47; S48

Original questions

Q1 · Mechanism · Drug Metabolism & CYP450 Interactions

A drug undergoes glucuronidation, which attaches a polar group to the molecule. Which statement best describes this step?

- **A** A Phase I oxidation that unmasks a reactive functional group on the drug
- **B** A CYP450-driven hydrolysis that always precedes any Phase II reaction
- **C** A Phase II reaction that makes the drug more lipophilic and longer-acting
- **D** A Phase II conjugation that makes the drug more water-soluble for excretion
- **E** A reduction reaction that converts a prodrug into its active form

ANSWER: D

Glucuronidation is a Phase II (conjugation) reaction that attaches a polar group, producing a more water-soluble product for renal/biliary excretion (S5). S5

- **A.** Oxidation is Phase I; glucuronidation is Phase II conjugation (S5).
- **B.** Hydrolysis is Phase I, and Phase I does not always come first (S5, S5 notes).
- **C.** Phase II produces a more water-soluble (polar) product (S5).
- **E.** Reduction is a Phase I reaction (S5).

Q2 · Clinical vignette · Drug Metabolism & CYP450 Interactions

A 19-year-old arrives 6 hours after a large intentional acetaminophen ingestion. Labs are pending. Which treatment best protects against fulminant liver injury?

- **A** Sodium bicarbonate to alkalinize the urine and speed excretion
- **B** N-acetylcysteine, started within the 8–16 hour window
- **C** Protamine to bind and neutralize the circulating drug
- **D** Rifampin to induce CYP enzymes and clear the drug faster
- **E** Waiting for levels before giving any treatment at all

ANSWER: B

N-acetylcysteine given within 8–16 hours protects against acetaminophen hepatotoxicity by replacing depleted glutathione (p. 66). p. 66

- **A.** Urine alkalinization is used for aspirin overdose, a weak acid (p. 20).
- **C.** Protamine neutralizes heparin (p. 26), not acetaminophen.
- **D.** Induction would increase toxic metabolite formation, not protect the liver (p. 66).
- **E.** The protective window is 8–16 hours, so delay risks liver failure (p. 66).

Q3 · Clinical vignette · Drug Metabolism & CYP450 Interactions

A heavy smoker on a CYP1A2-metabolized drug quits smoking abruptly. What should the PA anticipate?

- **A** No change, since smoking affects only the lungs
- **B** Drug levels will fall because metabolism speeds up
- **C** The drug becomes a prodrug that needs activation
- **D** Drug levels may rise as CYP1A induction fades
- **E** Renal clearance will double right away

ANSWER: D

Tobacco smoke induces CYP1A through AhR; when the inducer stops, metabolism slows and levels can rise (p. 60; p. 72). p. 60; p. 72

- **A.** Smoke chemicals induce liver CYP1A enzymes (p. 60).
- **B.** Quitting removes induction, so metabolism slows (p. 60).
- **C.** Quitting does not change how the drug is activated (p. 60).
- **E.** The effect is on hepatic metabolism, not renal clearance (p. 60).

Q4 · Clinical vignette · Drug Excretion & Renal Dosing

A patient arrives after an aspirin overdose, and the team alkalinizes the urine. How does this increase elimination of the drug?

- **A** It shifts elimination from the kidney to bile and feces
- **B** It makes the drug more lipophilic so it is reabsorbed
- **C** It increases glomerular filtration of protein-bound drug
- **D** It activates tubular transporters that secrete the drug
- **E** It traps ionized drug in the tubule, blocking its reabsorption

ANSWER: E

Urine pH manipulation works by trapping ionized drug in the tubule, blocking reabsorption (S14 notes); only lipophilic, un-ionized drug is reabsorbed (S14). S14; S14 (notes)

- **A.** The mechanism acts in the renal tubule (S14 notes).
- **B.** Lipophilic, un-ionized drug is reabsorbed — the opposite of the goal (S14).
- **C.** Only free (unbound) drug is filtered (S14).
- **D.** Secretion uses active transporters (S14), but urine pH works by blocking reabsorption (S14 notes).

Q5 · Mechanism · Drug Excretion & Renal Dosing

Which renal process REDUCES the net elimination of a drug?

- **A** Tubular secretion by active transporters
- **B** Passive reabsorption of lipophilic, un-ionized drug
- **C** Excretion of volatile agents by the lungs
- **D** Filtration of drug based on molecular size
- **E** Glomerular filtration of free, unbound drug

ANSWER: B

Tubular reabsorption: lipophilic, un-ionized drug passively diffuses back into the blood, reducing net elimination (S14). S14

- **A.** Secretion moves additional drug INTO the tubule (S14).
- **C.** Lung elimination removes volatile agents; it's not a renal process (S13).
- **D.** This is glomerular filtration, which adds drug to the tubule (S14).
- **E.** Filtration moves drug INTO the tubule (S14).

Q6 · Clinical vignette · PK Calculations: Half-Life, Steady State & Clearance

A patient receives an IV dose that gives a starting plasma concentration of 100 mg/L. The drug's half-life is 6 hours. What is the plasma concentration at 18 hours?

A	6.25 mg/L, with 93.75% of the drug eliminated
B	25 mg/L, with 75% of the drug eliminated
C	12.5 mg/L, with 87.5% of the drug eliminated
D	3.125 mg/L, with about 97% eliminated
E	50 mg/L, with 50% of the drug eliminated

ANSWER: C

18 h = 3 half-lives: C(t) = 100 × (1/2)^3 = 12.5 mg/L (87.5% eliminated) (S19). S19

- **A.** That is 24 h — 4 half-lives (S19).
- **B.** That is 12 h — 2 half-lives (S19).
- **D.** That is 30 h — 5 half-lives (S19).
- **E.** That is 6 h — 1 half-life (S19).

Q7 · Mechanism · PK Calculations: Half-Life, Steady State & Clearance

Which statement correctly pairs each dose type with the parameter it is based on?

A	Loading dose is based on bioavailability; maintenance dose on Vd
B	Loading dose is based on Vd; maintenance dose is based on clearance
C	Both loading and maintenance doses are based on clearance alone
D	Loading dose is based on clearance; maintenance dose is based on Vd
E	Both loading and maintenance doses are based on half-life alone

ANSWER: B

Loading doses are based on Vd; maintenance doses are based on clearance (S23 notes). S23 (notes)

- **A.** Loading = Vd; maintenance = clearance (S23 notes).
- **C.** Loading doses are based on Vd (S23 notes).
- **D.** This reverses the pairing (S23 notes).
- **E.** The lecture pairs loading with Vd and maintenance with clearance (S23 notes).

Q8 · Clinical vignette · PK Calculations: Half-Life, Steady State & Clearance

A drug is given every 8 hours instead of every 24 hours, keeping the same total daily dose. What happens to its average steady-state concentration?

- **A** It depends on Vd, which now changes
- **B** It triples because doses come more often
- **C** It falls because each dose is smaller
- **D** It stays the same; only peaks and troughs change
- **E** It doubles because half-life gets longer

ANSWER: D

The average steady-state level depends only on dosing rate and clearance; the interval changes only the swings (p. 51). p. 51

- **A.** Vd sets the loading dose, not the average level (p. 51; p. 52).
- **B.** Total dose per day is unchanged, so the average is unchanged (p. 51).
- **C.** Smaller, more frequent doses narrow the swings, not the average (p. 51).
- **E.** Dosing interval does not change half-life (p. 51).

Q9 · Clinical vignette · Pharmacogenomics

A patient is about to start azathioprine. Testing shows TPMT deficiency. What is the main clinical concern?

- **A** Reduced activation of the drug into its active form
- **B** Faster CYP2D6 metabolism of the drug and lost effect
- **C** Increased risk of toxicity from the thiopurine drug
- **D** Severe HLA-associated hypersensitivity to the drug
- **E** No concern, since genotype never affects therapy

ANSWER: C

TPMT deficiency raises toxicity risk with thiopurine drugs such as azathioprine (S33). S33

- **A.** Reduced activation is the CYP2C19–clopidogrel example (S33).
- **B.** CYP2D6 variation affects antidepressants and opioids (S33).
- **D.** That is the HLA allele example (e.g., HLA-B*5801 with allopurinol) (S33, S36).
- **E.** Genotype is one input among several — it still matters (S34).

Q10 · Mechanism · Pharmacogenomics

Which pairing of a genetic variant with the drug it most affects is correct?

- **A** CYP2C19 variation — allopurinol reactions
- **B** CYP2D6 variation — azathioprine toxicity
- **C** HLA-B*5801 — clopidogrel activation
- **D** CYP2C19 variation — clopidogrel activation
- **E** TPMT variation — codeine activation

ANSWER: D

CYP2C19 variation affects clopidogrel activation and some antidepressant/PPI dosing (S33). S33; S36

- **A.** Allopurinol reactions are HLA-B*5801 (S36).
- **B.** Azathioprine toxicity is TPMT deficiency (S33).
- **C.** HLA-B*5801 is tied to allopurinol hypersensitivity (S36).
- **E.** Codeine activation is CYP2D6 (S33).

Q11 · Mechanism · Pharmacogenomics

Which statement about using pharmacogenomic results best matches the lecture's personalized-pharmacotherapy model?

- **A** Genotype affects adverse-reaction risk but never drug metabolism
- **B** Genotype alone should determine both the drug and its dose
- **C** Genotype becomes irrelevant once a drug reaches steady state
- **D** Genotype matters only for drugs that are given by the IV route
- **E** Genotype is one input alongside phenotype, disease, and other drugs

ANSWER: E

Pharmacogenomics is one input among several — genotype alone never dictates therapy in isolation (S34). S32; S34

- **A.** Genetic variation can alter metabolism (e.g., CYP activity) (S32).
- **B.** Genotype alone never dictates therapy in isolation (S34).
- **C.** Nothing in the materials limits genotype's relevance this way (S32, S34).
- **D.** Genetic variation alters metabolism, transport, targets, and response (S32).

Q12 · Clinical vignette · Pharmacogenomics

A 62-year-old has a coronary stent placed. Genotyping shows he is a CYP2C19 poor metabolizer. Which antiplatelet choice does CPIC guidance support?

A Prasugrel or ticagrelor instead of clopidogrel

B Azathioprine at 30–70% of the usual starting dose

C Clopidogrel at the standard dose with routine follow-up

D Clopidogrel plus a CYP2C19 inhibitor to raise its level

E Codeine-free analgesia plus standard-dose clopidogrel

ANSWER: A

Clopidogrel needs CYP2C19 activation; poor metabolizers make little active drug, so prasugrel or ticagrelor is recommended (p. 81; p. 82). p. 81-82

- **B.** That dose rule is for TPMT-intermediate thiopurine use (p. 83).
- **C.** Standard clopidogrel risks stent thrombosis in poor metabolizers (p. 81).
- **D.** Inhibiting CYP2C19 would reduce clopidogrel activation further (p. 81).
- **E.** Codeine relates to CYP2D6, not antiplatelet choice (p. 81).

Q13 · Clinical vignette · Putting It All Together: Clinical Decision-Making

A patient starts a new drug that binds the same receptor as their home medication, and the home medication's effect becomes weaker. Raising the home medication's dose restores the full effect. What does this support?

- **A** The new drug is a strong CYP3A4 inducer
- **B** The new drug is an irreversible antagonist
- **C** The home medication has become toxic
- **D** The new drug is a competitive antagonist
- **E** The new drug is a full agonist itself

ANSWER: D

A competitive antagonist can be overcome by increasing the agonist dose — restoring the effect supports competitive rather than irreversible antagonism (S41). S41

- **A.** Nothing here involves CYP metabolism; it's receptor competition (S41).
- **B.** Irreversible antagonism would NOT be overcome by raising the dose (S41).
- **C.** A weaker effect restored by a higher dose points to competition (S41).
- **E.** A drug that weakens the home medication's effect is acting as an antagonist (S41).

Q14 · Clinical vignette · Putting It All Together: Clinical Decision-Making

A 68-year-old woman with stage 4 CKD (eGFR 24 mL/min) is starting a narrow-TI oral antiarrhythmic eliminated mainly unchanged by the kidney. Three weeks ago she began a strong CYP3A4 inhibitor. What is the best step before the prescription is filled?

- **A** Shorten the dosing interval so steady state is reached more quickly
- **B** Prescribe the standard dose, since only hepatic drugs are affected by CKD
- **C** Increase the dose, since the CYP3A4 inhibitor will lower the drug's level
- **D** Start the drug, and check renal function only if symptoms appear later
- **E** Check eGFR/CrCl, lower the dose or extend the interval, and plan close monitoring

ANSWER: E

Reduced renal clearance will raise the drug's concentration and prolong its half-life; before dispensing, check eGFR/CrCl, consider a lower dose or extended interval, and monitor renal function and toxicity (SG U1-S). SG U1-S; S15; S28

- **A.** Intervals are EXTENDED in renal impairment; shortening raises accumulation (S17, S28).
- **B.** CKD reduces renal clearance, raising concentration and half-life of renally-eliminated drugs (S15, SG U1-S).
- **C.** A CYP3A4 inhibitor RAISES substrate levels (S6); induction lowers them.
- **D.** Check eGFR/CrCl and adjust BEFORE dispensing, then monitor closely (SG U1-S).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO1.1** Define and describe fundamental pharmacologic concepts, including pharmacokinetics, pharmacodynamics, drug-receptor interactions, dose-response relationships, therapeutic index, half-life, and steady state.
- [ ] **IO1.2** Describe the processes of drug absorption, distribution, metabolism, and excretion and factors that influence these processes.
- [ ] **IO1.3** Explain how patient-specific factors, including age, pregnancy, renal and hepatic function, and pharmacogenetic variation, may affect medication response.
- [ ] **IO1.4** Identify common types and mechanisms of adverse drug reactions, drug-drug interactions, and drug-disease interactions.
- [ ] **IO1.5** Interpret common components of medication orders and prescriptions, including drug name, dose, route, frequency, and duration.

- [ ] **SO1** Explain drug biotransformation (Phase I/II metabolism) and the clinical impact of CYP450 enzyme inhibition and induction.
- [ ] **SO2** Explain mechanisms of drug excretion and how renal function affects dosing.
- [ ] **SO3** Calculate and apply basic pharmacokinetic parameters (half-life, steady state, clearance).
- [ ] **SO4** Describe how pharmacogenomic variation can alter drug response.
- [ ] **SO5** Apply a structured framework to clinical prescribing decisions.